Experiment 8

AIM: Use Postman to test sample API.

1. Objective

- Understand how to test REST APIs using Postman by sending different types of HTTP requests.
    - Get
    - Post
    - Put
    - Delete

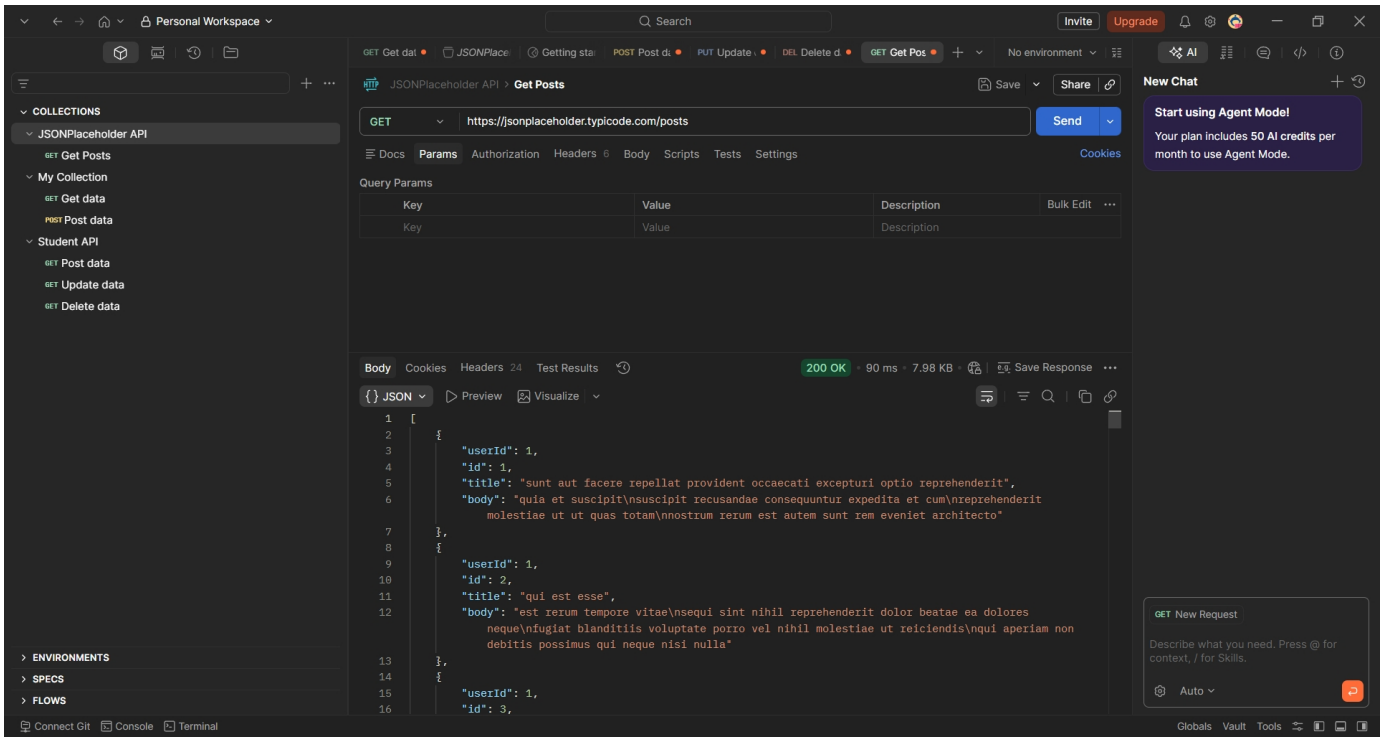
2. Prerequisites

- postman
- API to be tested
    - Create a simple node.js API / use some free fake API

3. Test API

Get:

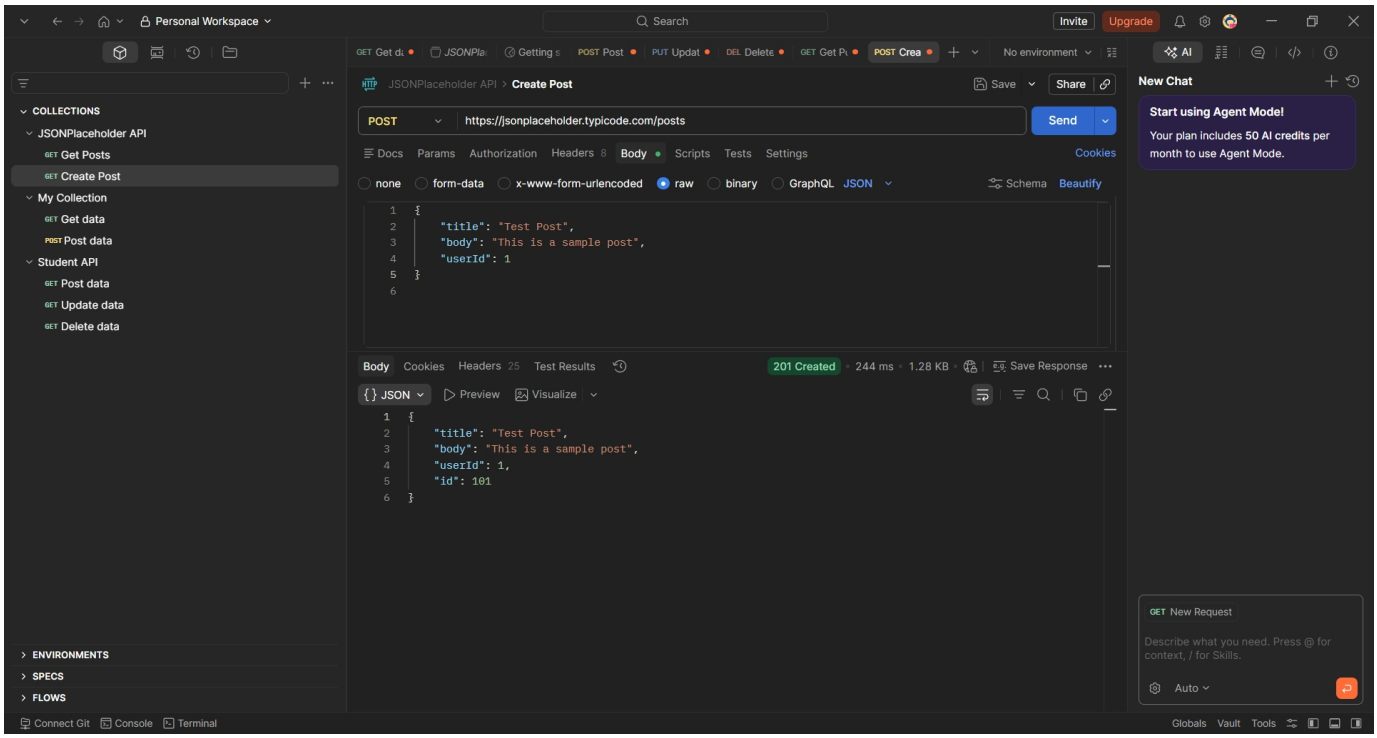
URL: https://jsonplaceholder.typicode.com/posts



Post:

https://jsonplaceholder.typicode.com/posts

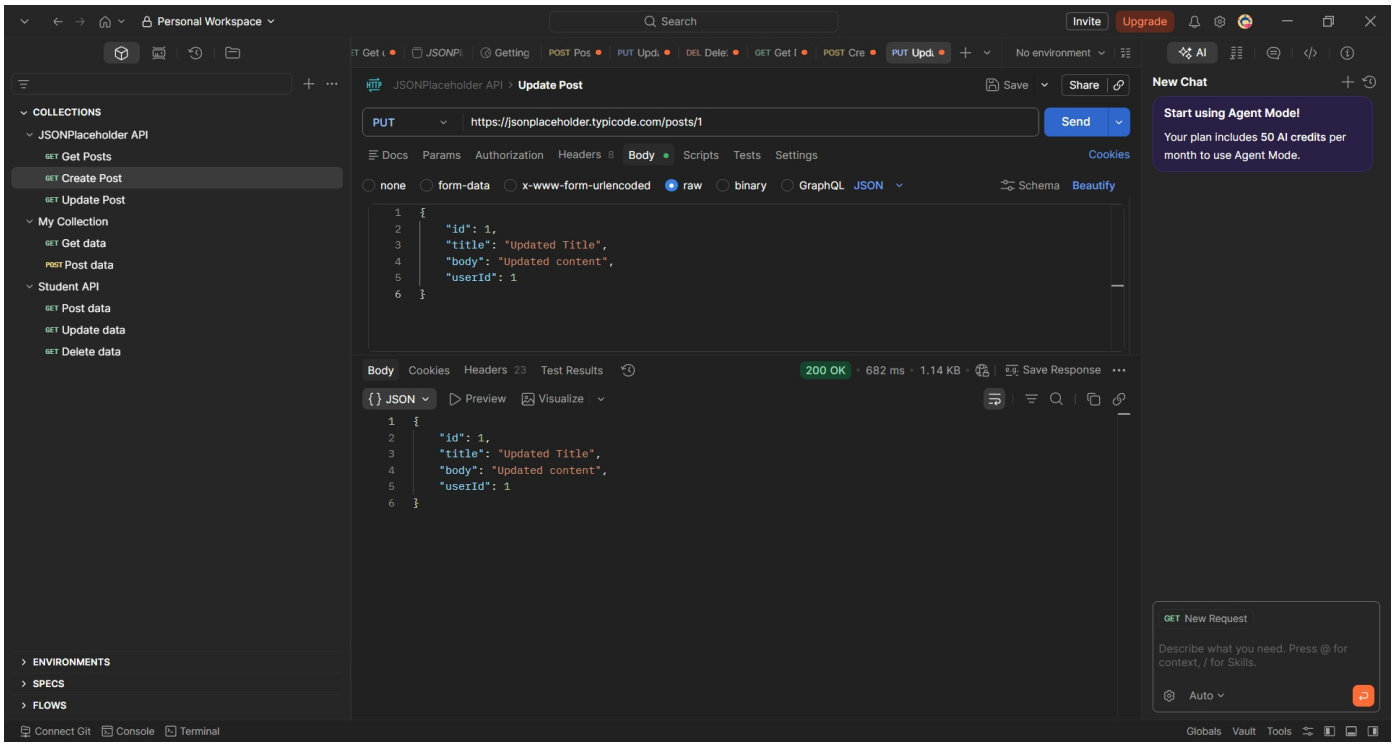
```
{
"title": "Test Post",
"body": "This is a sample post",
"userId": 1
}
```



Put:

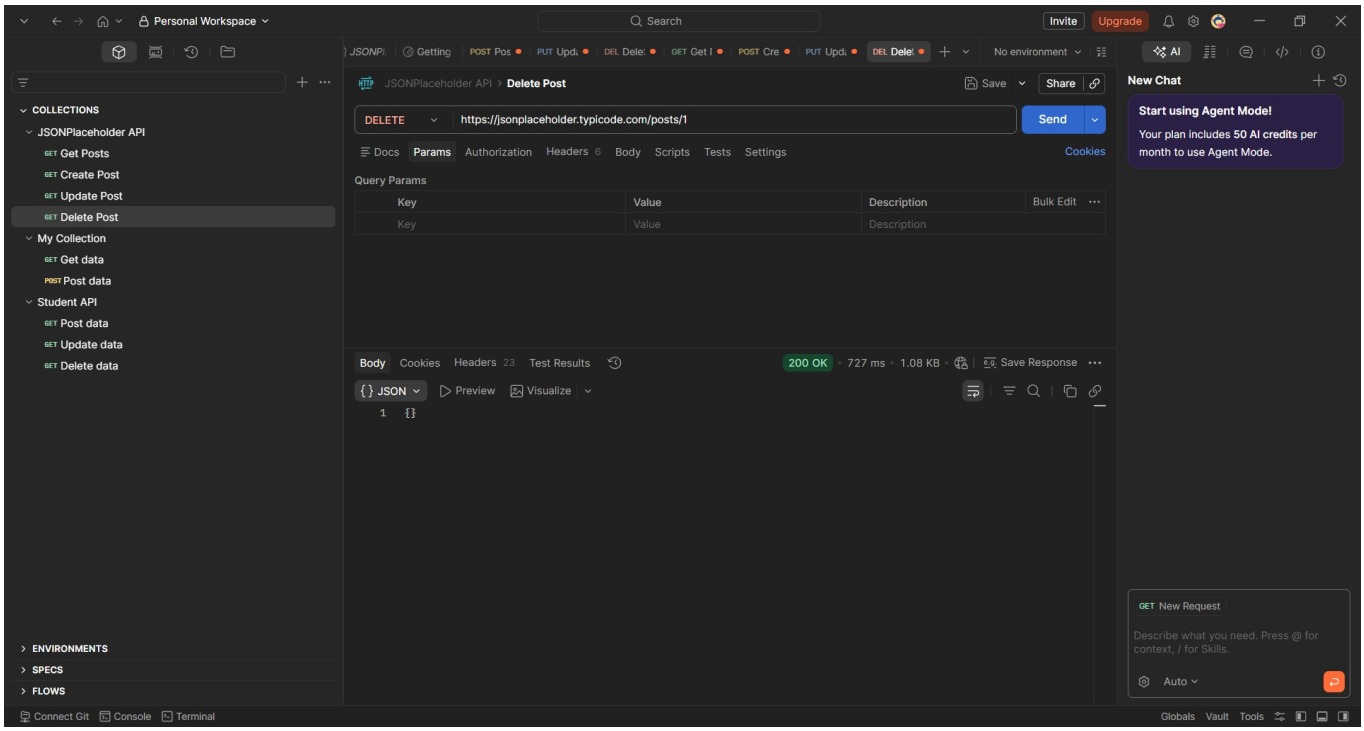
https://jsonplaceholder.typicode.com/posts/1

```
{
"id": 1,
"title": "Updated Title",
"body": "Updated content",
"userId": 1
}
```



Delete:

https://jsonplaceholder.typicode.com/posts/1



4. Create node.js API

Step 1: Create Project

```
mkdir simple-api

cd simple-api

npm init -y

npm install express
```

Step 2: Create server.js file

```
const express = require('express');

const app = express();

app.use(express.json());
```

```
let students = [
{ id: 1, name: "Viraj", branch: "CE" },
{ id: 2, name: "Ravi", branch: "CE" }
];
app.get('/students', (req, res) => {
res.json(students);
});
app.get('/students/:id', (req, res) => {
const student = students.find(s => s.id == req.params.id);
res.json(student);
});
app.post('/students', (req, res) => {
const newStudent = {
id: students.length + 1,
name: req.body.name,
branch: req.body.branch
};
students.push(newStudent);
res.status(201).json(newStudent);
});
app.put('/students/:id', (req, res) => {
const student = students.find(s => s.id == req.params.id);
if (student) {
student.name = req.body.name;
student.branch = req.body.branch;
```

```
res.json(student);
} else {
res.status(404).json({ message: "Student not found" });
}
});
// DELETE - Remove student
app.delete('/students/:id', (req, res) => {
students = students.filter(s => s.id != req.params.id);
res.json({ message: "Student deleted" });
});
app.listen(3000, () => {
console.log("Server running on http://localhost:3000");
});
```

Step 3: Run server

Node server.js

Command :

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\prash>cd C:\Users\prash\Desktop
C:\Users\prash\Desktop>mkdir simple-api
C:\Users\prash\Desktop>cd simple-api
C:\Users\prash\Desktop\simple-api>npm init -y
Wrote to C:\Users\prash\Desktop\simple-api\package.json:
{
```

```
  "name": "simple-api",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

C:\Users\prash\Desktop\simple-api>npm install express

added 65 packages, and audited 66 packages in 2s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
C:\Users\prash\Desktop\simple-api>notepad server.js
Server running on http://localhost:3000
```

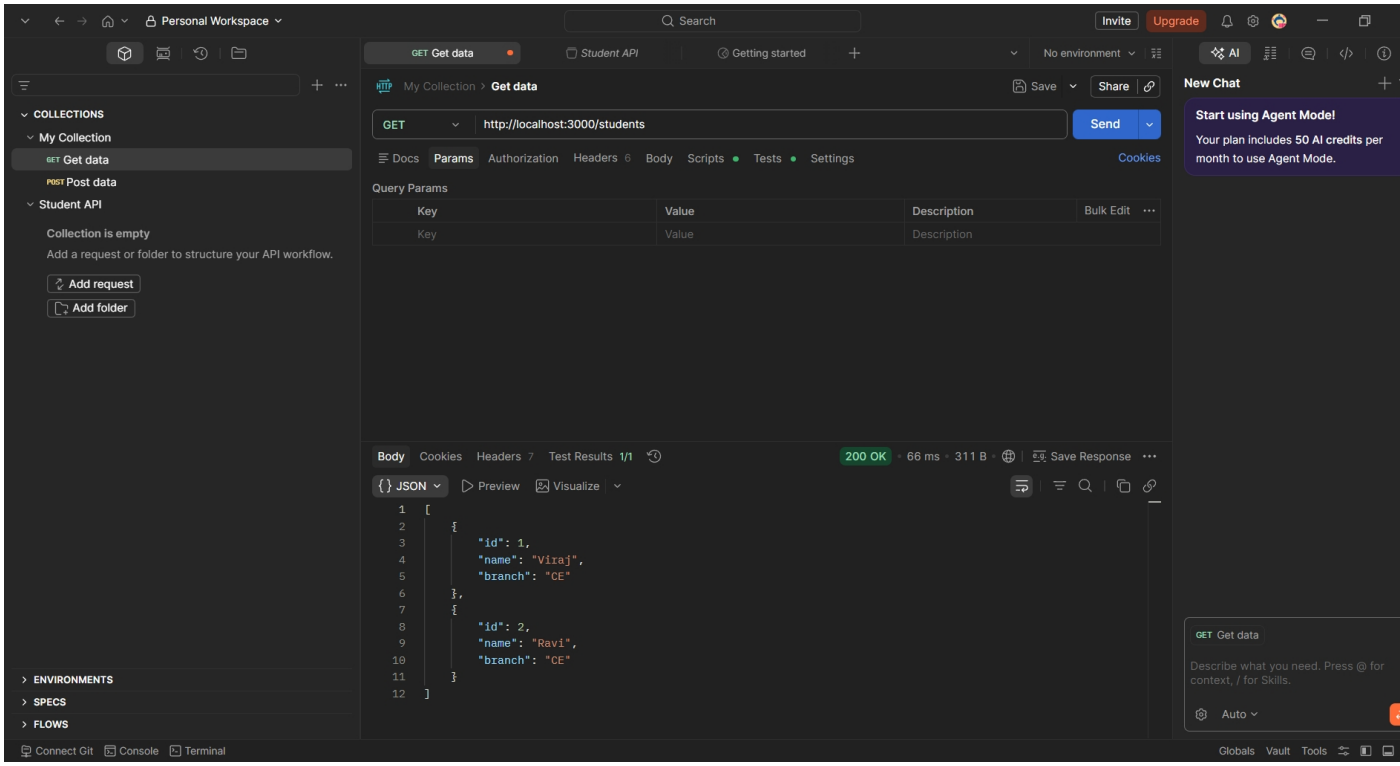
5. Procedure TEST using postman

1. GetAll:

Type: Get

URL : http://localhost:3000/students

Check response



2. Post:

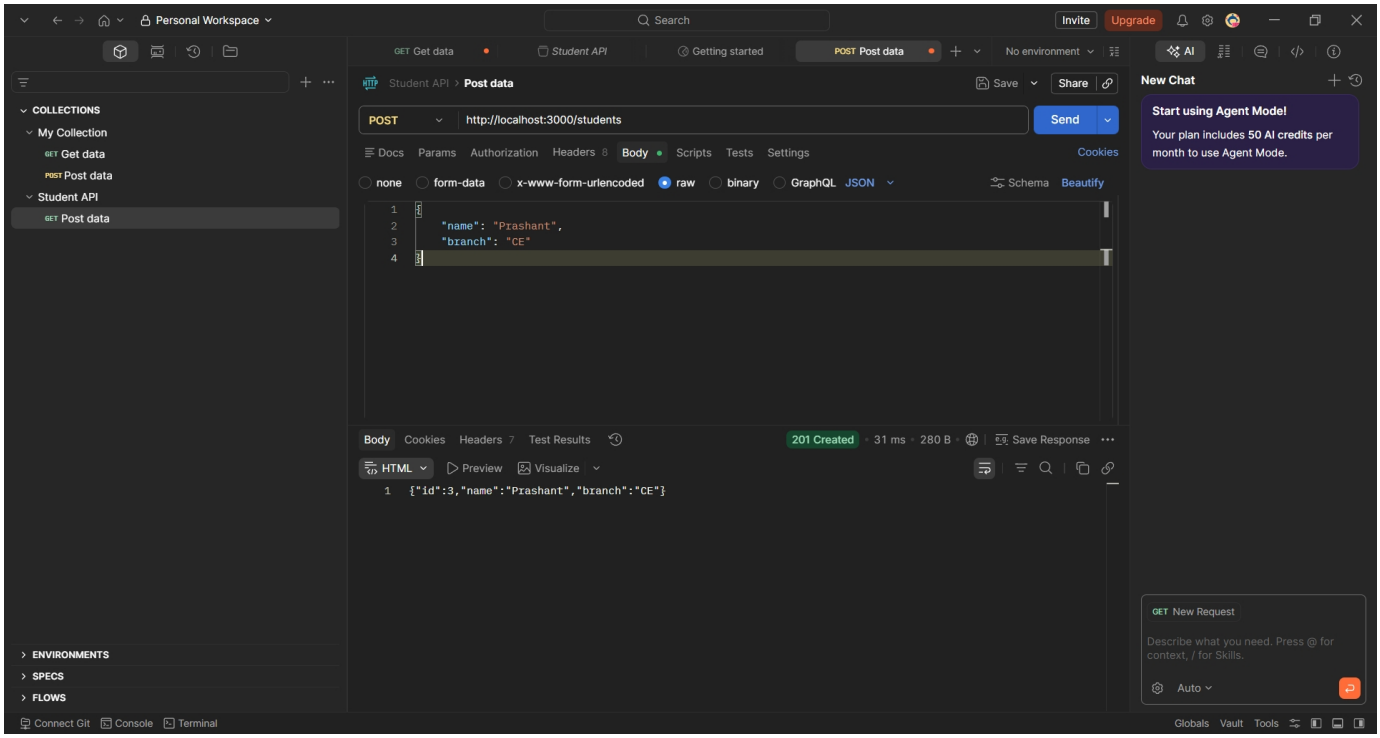
Type: Post

URL: http://localhost:3000/students

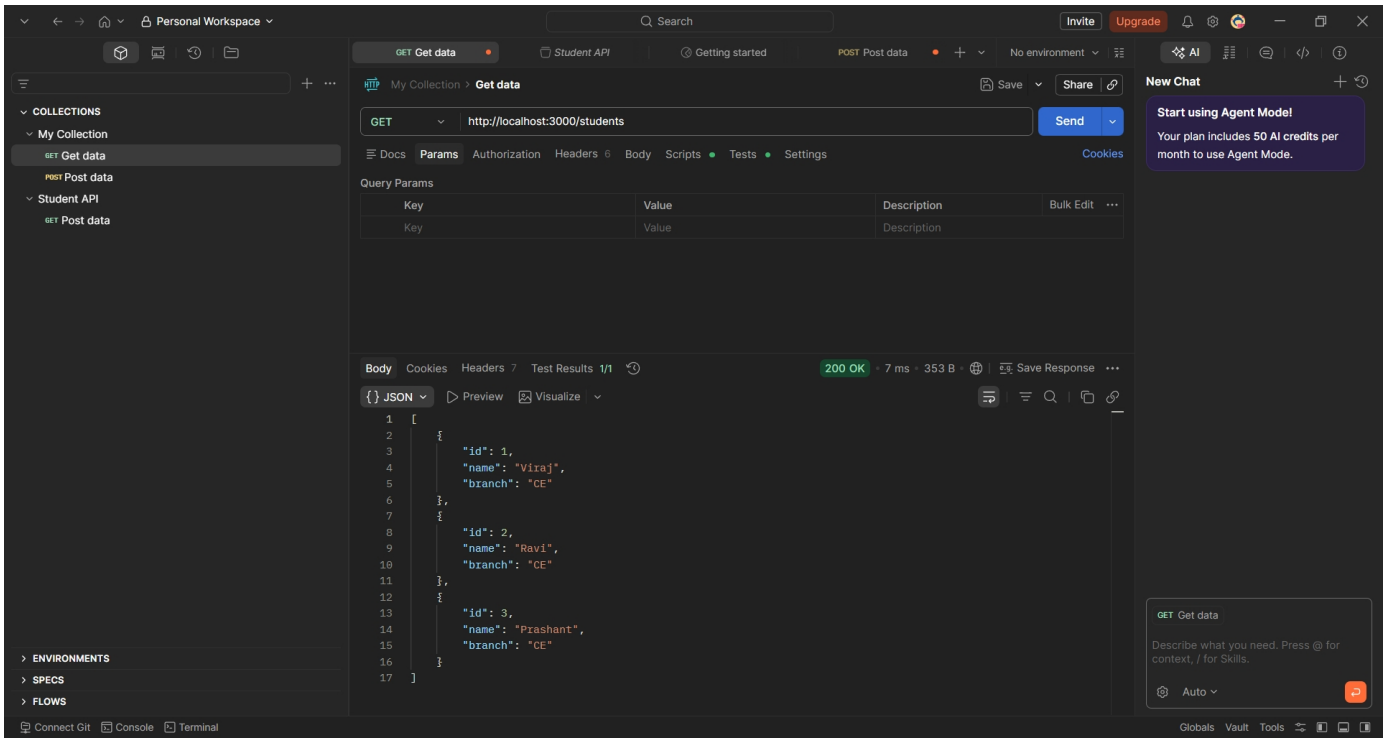
Body:

```
{
"name": "Prashant",
"branch": "CE"
}
```

Check response



Send Get request to see new data inserted



3. Put request:

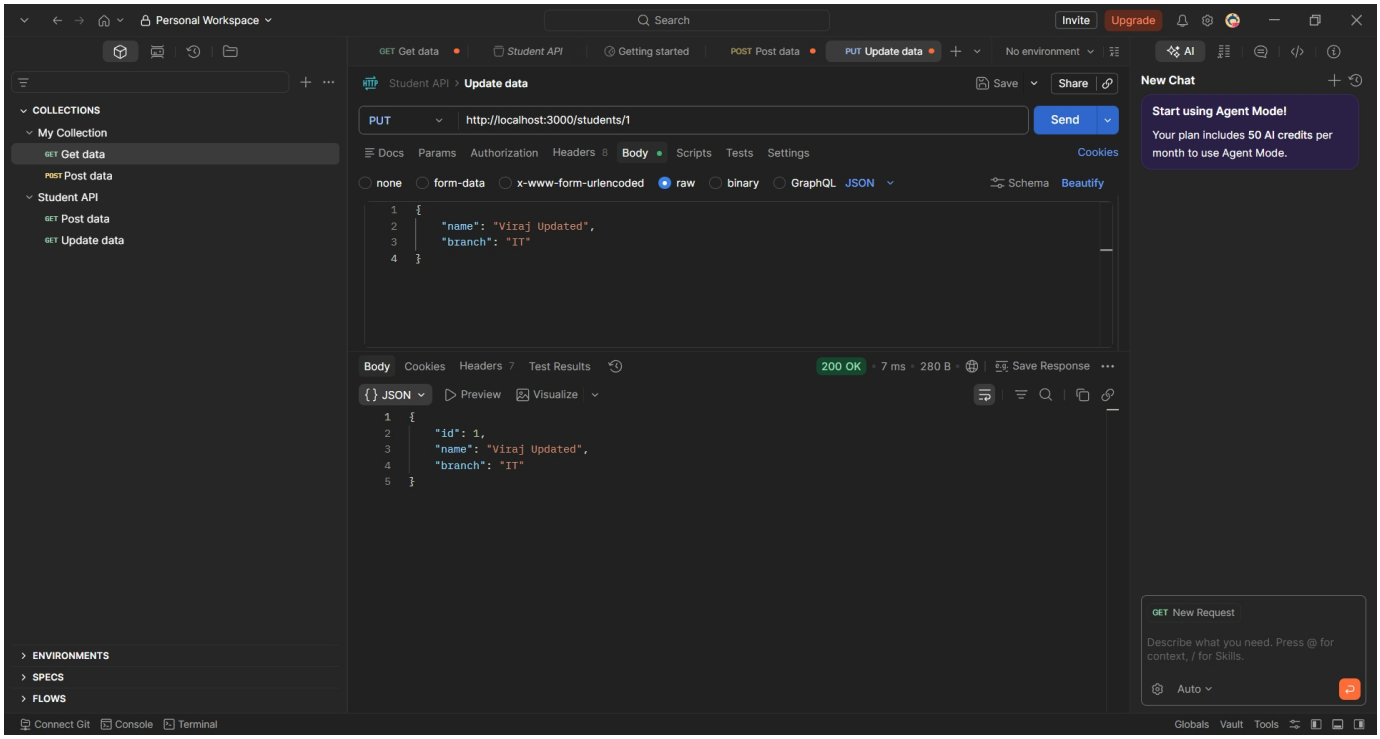
Type: PUT

URL : http://localhost:3000/students/1

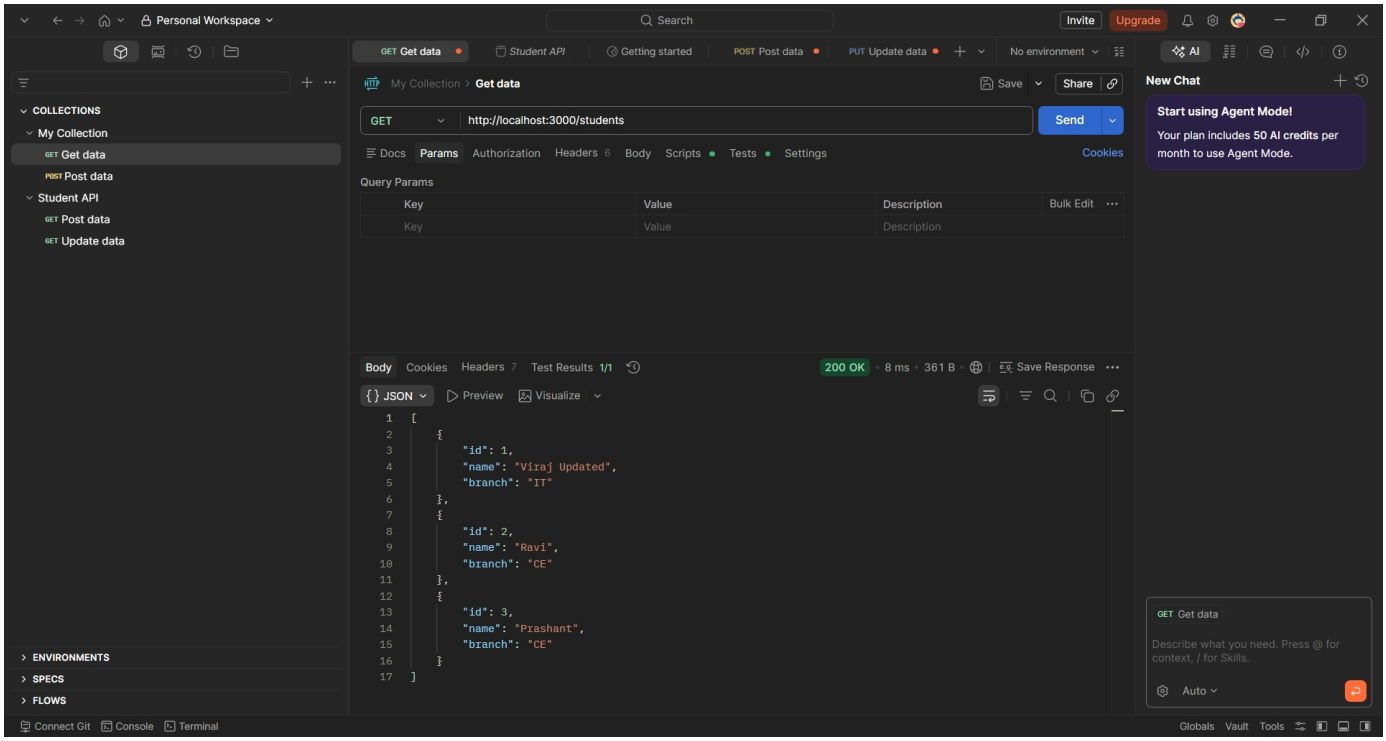
Body:

```
{
"name": "Viraj Updated",
"branch": "IT"
}
```

Check response



Send Get request to see updated data

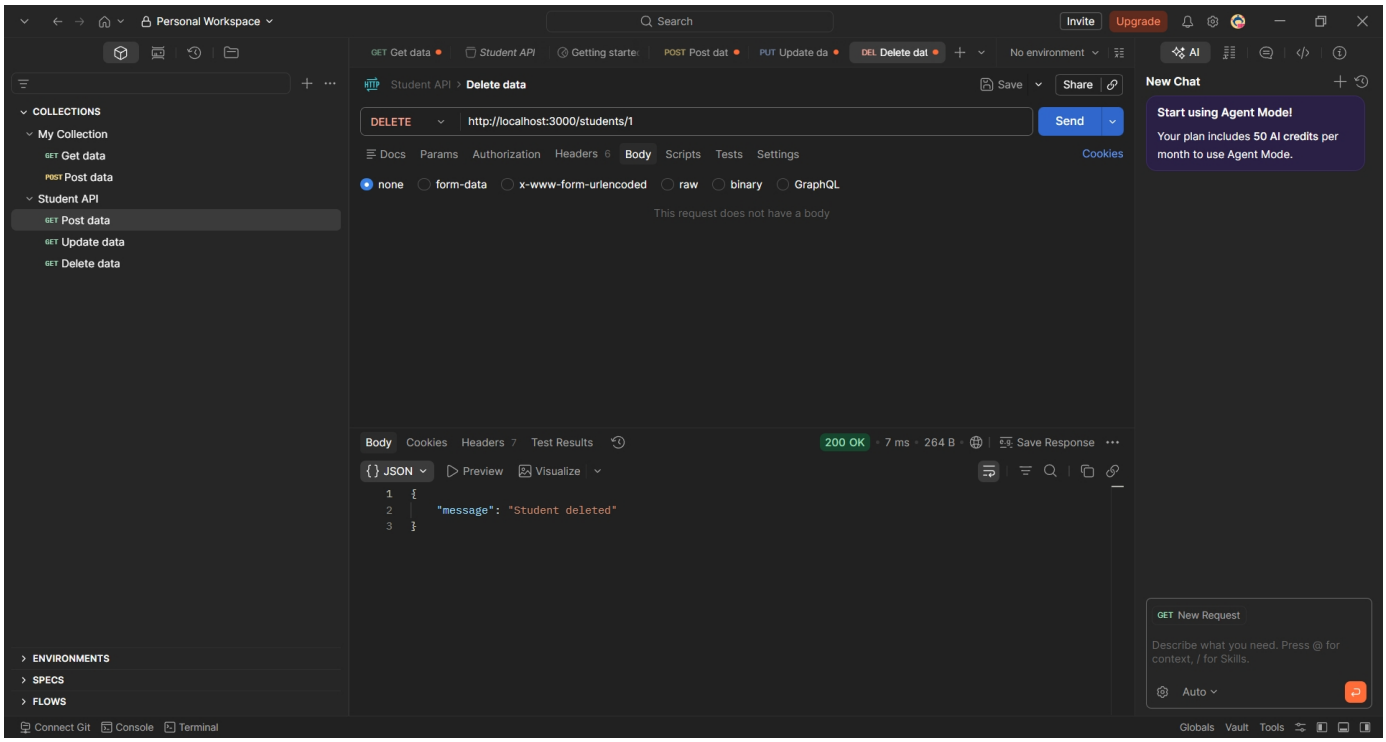


4. Delete request:

Type: DELETE

URL : http://localhost:3000/students/1

Check response



Send Get request to see updated data